



CERTIFIED MATERIAL TEST REPORT
CERTIFICATE OF COMPLIANCE
CERTIFICATE # 251543
REVISION 0

Tube Forgings of America, Inc.
5200 N.W. Front Avenue
Portland, OR 97210
ISO 9001:2015 Certified
7/8/2026

Sold To	DNOW LP #MI04 1701 Gover Pkwy. Mt Pleasant, MI 48858-8140	Ship To	DNOW LP #MI04 1701 Gover Pkwy. Mt Pleasant, MI 48858-8140
Customer ID	34217	Heat Code	A3124 ✓
TFA Sales Order	3310682-1	Grade	WPB
Purchase Order	663014514	Raw Material Heat	4216776DF
TFA Part ID	15190	Process	WPB-CN31
Description	4 X 2 XH TEE	Mill	CHANGBAO
Quantity	6		6 ea.
Customer Part ID	70791		

Austenitized @:	1670° F	Tempered @:	°	Stress Relieved @:		Medium:
Held For:	60 mins/min	Held For:		Held For:		

Heat Analysis Chemistry

C	Mn	P	S	Si	Mo	Cr	Ni	Cu	V	Cb	B	Ti	Al	N	Co	Ca	C.E.
0.17	0.82	0.009	0.006	0.26	0.00	0.03	0.01	0.02	0.00								0.31

Tensile Test Results

Test	Name	Yield	Tensile	% Elong	R/A %	Gage Length	HBW	In-Process Test 1	In-Process Test 2	Tensile Orientation	Test Bar Type & Size
1	BASE	48732 (psi)	70343 (psi)	37.00		2"	137	137	137	LONGITUDINAL	STRIP

Charpy Impact Test Results

	Impact Energy				Lateral Expansion (Mils)			% Shear				Test Temp.	Size (mm)	Charpy Orientation
	1	2	3	Avg	1	2	3	1	2	3	Avg			

Notes/Comments

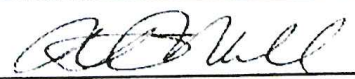
1. THESE FITTINGS WERE COLD FORMED AND NORMALIZED AT 1670F AND AIR COOLED. 2. THESE FITTINGS WERE S2 TENSILE TESTED TO VERIFY PHYSICAL PROPERTIES AFTER THE HEAT TREAT PROCEDURE. 3. THE STARTING MATERIAL IS ASTM A106 GRB SEAMLESS PIPE. 4. THE HARDNESS DOES NOT EXCEED 197BHN PART TO PART. 5. THIS PRODUCT IS SEAMLESS. 6. TEES WERE "MAG PARTICLE INSPECTED IN ACCORDANCE WITH (ASTM / ASME - SA / A234) REQUIREMENTS

1) MATERIAL IS MANUFACTURED TO ASTM A234-24 GR WPB, ALSO PER ASME SEC II SA234 GR WPB-2023 EDITION. 2) DESIGN AND DIMENSIONS TO ASME B16.9-2018. 3) MATERIAL MEETS ANSI/NACE MR0175/ISO15156:2015 AND ANSI/ NACE MR0103/ISO17945:2015. 4) MANUFACTURED IN AN ISO 9001 REGISTERED FACILITY, CERTIFICATE #: 30248. 5) TFA, INC. ORGANIZATIONAL STRUCTURE AND CERTIFICATION COMPLIES WITH BS EN 10204-2004 TYPE 3.1.B. 6) TFA INC. FITTINGS SATISFY THE MATERIAL REQUIREMENT OF PRESSURE EQUIPMENT DIRECTIVE (PED) 2014/68/EU, ANNEX 1, PARAGRAPH 4.3. 7) FITTINGS ARE MANUFACTURED IN THE USA. RAW MATERIAL IS FROM DOMESTIC AND GLOBAL SOURCES. 8) FITTINGS WERE MANUFACTURED, SAMPLED, TESTED AND INSPECTED IN ACCORDANCE WITH THE ORDER AND SPECIFICATIONS AND WERE FOUND TO MEET THE REQUIREMENTS.

*UB = Unbroken Test

Heat: A3124 Grade: WPB

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Scot Marshall, Manager Quality Assurance

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